

Deep Learning Project for Beginners part 1

Overview

If you are an experienced person or a fresher in the field of data science, you must have already heard about the term Deep learning. Deep learning takes its inspiration from the structure and function of the brain. In simple terms, deep learning tries to draw conclusions similar to humans if given a logical structure to analyze the data. Self-driving cars are phenomenal examples of deep learning. As the world is moving towards automation it becomes important to understand the key concepts behind it. In this series of projects, we try to give you basic ideas related to deep learning and its different algorithms.

Aim

To introduce the Deep Neural Network and its implementation

Data Description

The dataset used is a licensed dataset. It contains information about 86K different businesses over various features. The target variable is the status of license which has five different categories.

Tech Stack

- Language: Python
- Libraries: pandas, seaborn, numpy, matplotlib, scikit learn, h2o, tensorflow

Approach

1. Data Description
2. Exploratory Data Analysis
3. Data Cleaning
4. Preparing data for analysis
5. Base model building using h2o
6. Building deep neural network model
7. Predictions on test data
8. Model deployment using flask gunicorn

Modular code overview

input

- |_License_Data.csv
- |_test_data.csv

output

- |_dnn-model
- |_columns.mapping

src

- |_init__.py
- |_Engine.py
- |_ML_pipeline
 - |_init.py_.py
 - |_deploy.py
 - |_Predict.py
 - |_Preprocess.py
 - |_Train_Model.py
 - |_Utils.py
 - |_wsgi.py

Deep-Neural-Network.ipynb

Model_Api.ipynb

requirements.txt

Once you unzip the modular_code.zip file, you can find the following folders within it.

1. input
 2. src
 3. Output
 4. Deep-Neural-Network.ipynb
 5. Model_Api.ipynb
 6. requirements.txt
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1. The input folder contains the data that we have for analysis. In our case, it contains Licence_Data.csv and test_data.csv.

2. The src folder is the heart of the project. This folder contains all the modularized code for all the above steps in a modularized manner. It further includes the following.
 - a. ML_pipeline
 - b. engine.py

The ML_pipeline is a folder that contains all the functions put into different python files, which are appropriately named. These python functions are then called inside the Engine.py file.

3. The output folder contains the trained model and columns mapping which can be directly loaded for future use.
4. The requirements.txt file has all the required libraries with respective versions. Kindly install the file by using the command **pip install -r requirements.txt**
5. Deep-Neural-Network.ipynb and Model_Api.ipynb are the ipython notebooks we saw in the videos.

Project Takeaways

1. What is a Deep Neural Network?
2. Building blocks of Deep Neural Network
3. What is the Activation Function?
4. What is Feedforward?
5. What is Backpropagation?
6. Loss function and its examples
7. What is Dropout regularization?
8. Deep learning libraries such as Tensorflow, Pytorch, Pytorch lightning, Horovod
9. Understanding the Business context and objective
10. Data Cleaning
11. How to prepare data for modeling?
12. How to use the h2o framework for baseline modeling?
13. How to build a Deep Neural Network Model?
14. Hyperparameter tuning
15. Model predictions on test data
16. Model deployment using flask gunicorn
17. Predictions using the deployed model on the server